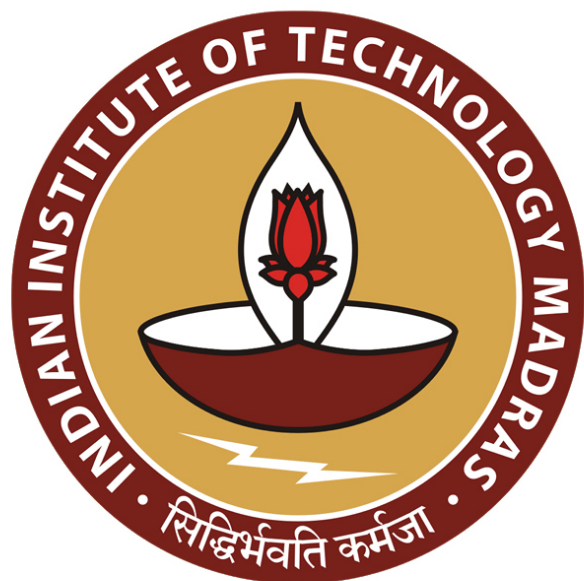


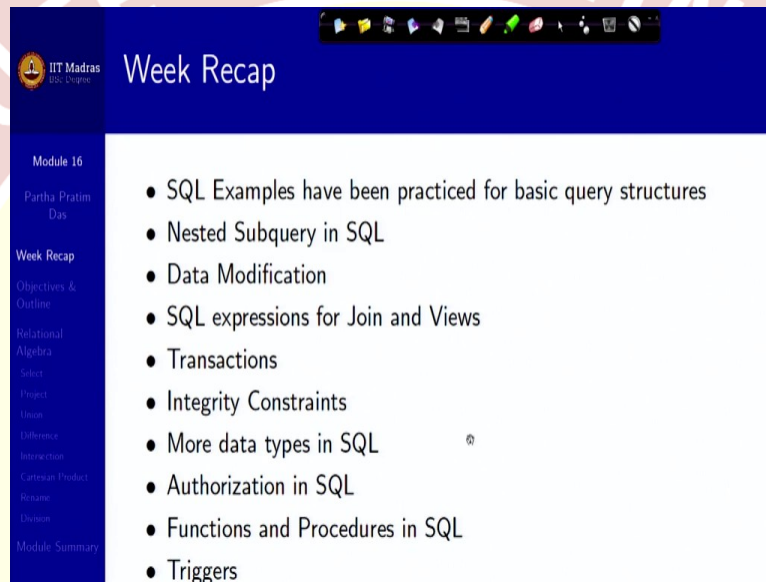


IIT Madras
ONLINE DEGREE

Database Management Systems
Professor Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Formal Relational Query Languages/1

Welcome to week 4, module 16 of Database Management Systems course in the IIT Madras online B.Sc program.

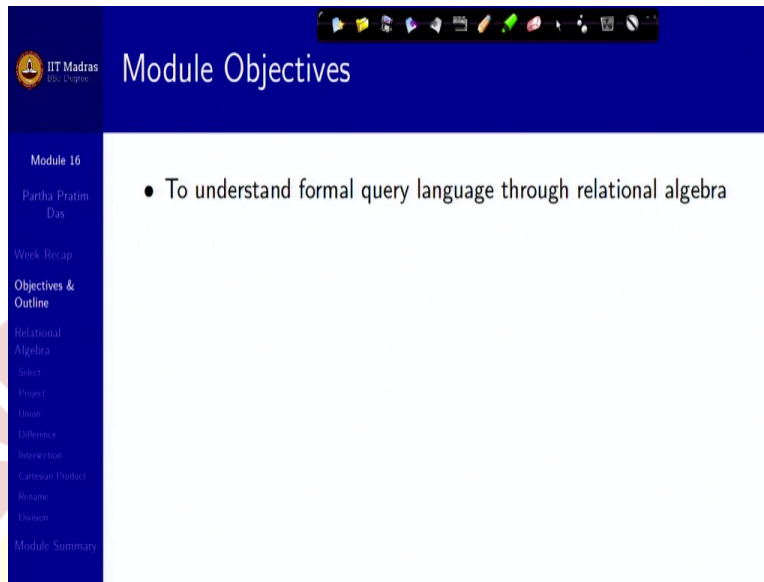
(Refer Slide Time: 0:30)



In the week last, we covered a lot of features of the SQL language starting from practice of the basic query structure to nested queries and data modification and joins, views, transactions integrity constraints, data types authorization and finally, also dealt with some advanced features of SQL which deal with the procedural interface of the procedural components in SQL functions and procedures and finally, discussed triggers in SQL.

So, with that, in week 3, we have kind of concluded the discussion on the SQL language for now, maybe some specific aspects will take up later on, but we will now slowly move on to going towards the modeling of the database and towards the design of the databases.

(Refer Slide Time: 1:39)



IIT Madras
100th Anniversary

Module Objectives

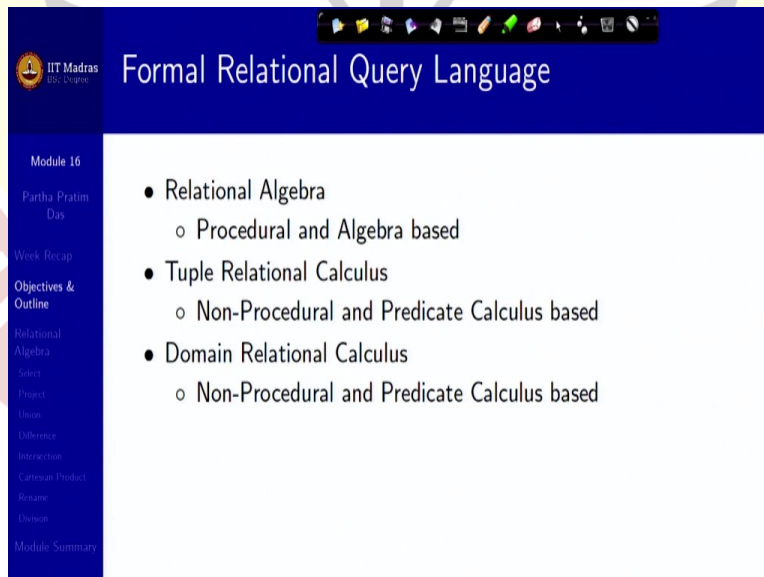
Module 16
Partha Pratim Das

Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Union
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

- To understand formal query language through relational algebra

As a foundational part of that, we would like to understand the formal query languages through relational algebra. We did talk about this a bit quite a bit actually earlier too, but we will take a little deeper look in a slow space and slowly develop on those models for being able to make designs later on.

(Refer Slide Time: 2:06)



IIT Madras
100th Anniversary

Formal Relational Query Language

Module 16
Partha Pratim Das

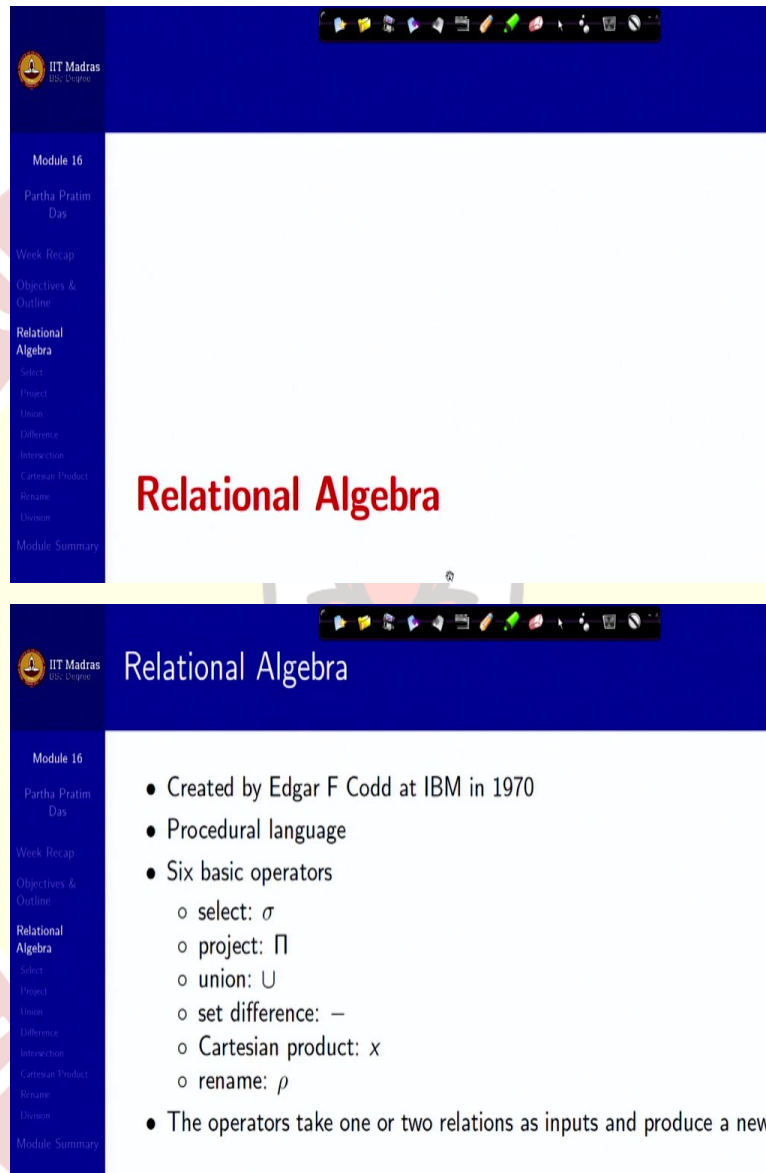
Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Union
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

- Relational Algebra
 - Procedural and Algebra based
- Tuple Relational Calculus
 - Non-Procedural and Predicate Calculus based
- Domain Relational Calculus
 - Non-Procedural and Predicate Calculus based

So, in this module, we talk about relational algebra. Overall as I said that there are 3 formal relational query language systems relational algebra, which is procedural algebra based and there

are 2 calculi, tuple relational calculus and domain relational calculus both of which are non-procedural and predicate calculus based.

(Refer Slide Time: 2:31)



The image displays two screenshots of a presentation slide titled "Relational Algebra" from IIT Madras. The top screenshot shows the title slide with a blue header and a sidebar menu. The bottom screenshot shows the same slide with a list of bullet points describing Relational Algebra.

Relational Algebra

- Created by Edgar F Codd at IBM in 1970
- Procedural language
- Six basic operators
 - select: σ
 - project: Π
 - union: $\cup$
 - set difference: $-$
 - Cartesian product: $\times$
 - rename: ρ
- The operators take one or two relations as inputs and produce a new

Now, for relational algebra, we have relations and this was created by Edgar Codd in IBM in 1970 is a procedural language it has to write relational algebra expressions to mean which shows the operations to be done, not it is not declarative in that way, it does not describe the result, but it describes a processes or operations that you do with the relations with using the six operators listed here. And the operators are unary or binary and it will always result in a new relation.

(Refer Slide Time: 3:19)

IIT Madras
1959

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Joins

Division

Module Summary

Select Operation

- Notation: $\sigma_p(r)$
- p is called the **selection predicate**
- Defined as:

$$\sigma_p(r) = \{t \mid t \in r \text{ and } p(t)\}$$

where p is a formula in propositional calculus consisting of **terms** connected by : $\wedge$ (**and**), $\vee$ (**or**), $\neg$ (**not**)

Each **term** is one of:

$$\langle \text{attribute} \rangle \text{ op } \langle \text{attribute} \rangle \text{ or } \langle \text{constant} \rangle$$

where op is one of: $=, \neq, >, \geq, <, \leq$

A
α
α
β
β

A
α
β

So, you know by now, the semantic meaning of these operations, the select operation, the notation is as you can, as you have already seen, where p this condition is called the selection predicate or the condition on which the selection is happening. So, if I write in set theoretic notation, which is what we will need more as I go into the calculus.

Then we have the collection of tuples t such that t belongs to r , it is a part of the relation and it satisfies this predicate $p(t)$, there could be different types of conjunction, disjunction, negation and those different components could be attributes or constants and comparison can be done in. So, this kind of example, you already know.

Here is another example showing you based on the data of $A=B$, $A=B$, this being these 3 and $D>5$, $D>5$, I think this example we had seen earlier also. So, that retains this one and this one. So, this is a select operation. Now, you have to start, you know understanding and practicing it from an algebraic mindset because we are going to use this to build more and more on this algebra.

(Refer Slide Time: 5:05)

IIT Madras
UGC Centre

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Project

Union

Difference

Intersection

Cartesian Product

Join

Division

Module Summary

Project Operation

- Notation: $\Pi_{A_1, A_2, \dots, A_k}(r)$
where A_1, A_2 are attribute names and r is a relation
- The result is defined as the relation of k columns obtained by erasing the columns that are not listed
- Duplicate rows removed from result, since relations are sets
- Example: To eliminate the *dept_name* attribute of *instructor*

$$\Pi_{ID, name, salary}(instructor)$$

A	I
α	1
α	2
β	3
β	4

A	C
α	1
α	1
β	1
β	2

Project operation which takes out a set of specified columns, it is very simple, but certainly when it takes out columns, it is possible that some tuples will become, will occur more than once they will be duplicates. And since every relation is a set now not a multiset like SQL, so the duplicates will be removed in every relational algebra operation. So, this is a project operation.

(Refer Slide Time: 5:35)

IIT Madras
UGC Centre

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Join

Union

Difference

Intersection

Cartesian Product

Join

Division

Module Summary

Union Operation

- Notation: $r \cup s$
- Defined as: $r \cup s = \{t \mid t \in r \text{ or } t \in s\}$
- For $r \cup s$ to be valid.
 - r, s must have the same arity (same number of attributes)
 - The attribute domains must be compatible (example: 2nd column of r deals with the same type of values as does the 2nd column of s)
 - Example: to find all courses taught in the Fall 2009 semester, or in the Spring 2010 semester, or in both

$$\Pi_{course_id}(\sigma_{semester="Fall"} \wedge year=2009(section)) \cup \Pi_{course_id}(\sigma_{semester="Spring"} \wedge year=2010(section))$$

A	B
α	1
α	2
β	1

r
 $\downarrow$

Then we have union operation which is simple union of two sets, $r \cup s$ is t belongs to r or t belongs to s and this is an inclusive or, which means that it can belong to both, but for the union to be defined the conditions are, that both of them must have the same arity that is the same

number of columns or same number of attributes and columns must be compatible, which means that the corresponding column in r and s must be of the same type, one column is integer other column is varchar, I cannot take a union.

That is not that will not be admissible. So, here is an example, which you have seen several times. So, finding out courses taught in Fall 2009 semester as well as Spring 2010 semester. So, you take out courses in Fall 2009, take out the `course_id` from that courses in Spring 2010 take out the `course_id`, do a union there is only one arity only one `course_id` the same type to a union we will get that result, a simple operation.

(Refer Slide Time: 6:56)




IIT Madras
Module 16
Partha Pratim Das
Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Union
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

Difference Operation

- Notation $r - s$
- Defined as: $r - s = \{t \mid t \in r \text{ and } t \notin s\}$
- Set differences must be taken between **compatible** relations
 - r and s must have the **same** arity
 - attribute domains of r and s must be compatible
- Example: to find all courses taught in the Fall 2009 semester, but not in the Spring 2010 semester

$$\Pi_{course_id}(\sigma_{semester="Fall" \wedge year=2009}(section)) - \Pi_{course_id}(\sigma_{semester="Spring" \wedge year=2010}(section))$$

A	B
α	1
α	2
β	1

r

A
α
β

s

Naturally, the other oversight operations which you have already seen is difference, so that t belongs to r and t does not belong to s , the same compatibility is required without that we cannot work in this case. So, example is to find all courses taught in Fall 2009 but not in semester Spring 2010.

(Refer Slide Time: 7:26)



Intersection Operation

Module 16
Partha Pratim Das
Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Union
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

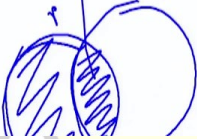
- Notation: $r \cap s$
- Defined as:
$$r \cap s = \{t | t \in r \text{ and } t \in s\}$$
- Assume:
 - r, s have the same arity
 - attributes of r and s are compatible
- Note: $r \cap s = r - (r - s)$

A	B
α	1
α	2
β	1

r

A
α

$r \cap s$



Intersection operation is finding out those tuples which belong to both. So, again the conditions is, they must have the same arity, they must be compatible in terms of type and we may note that intersection is expressible in terms of difference. So, I have r , I have s , this is r , this is s . So, I can do $r - s$, $r - s$ is this part then I do r minus this part.

So, from r I take out this part. So, from this entire of r I take out this part. So, what I am left with is this part, which is $r \cap s$. So, basically the point is that they are not independent operations one can be expressed in terms of the other, but having both helps in terms of writing queries quickly.

(Refer Slide Time: 8:30)

IIT Madras
Using Chrome

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Rename

Division

Module Summary

Cartesian-Product Operation

- Notation $r \times s$
- Defined as:

$$r \times s = \{t \mid t \in r \text{ and } q \in s\}$$
- Assume that attributes of $r(R)$ and $s(S)$ are disjoint. (That is, $R \cap S = \phi$)
- If attributes of $r(R)$ and $s(S)$ are not disjoint, then renaming must be used

A	B
α	1
β	2

A	B	C	D
α	1	α	1
α	1	β	1
α	1	β	2
α	1	γ	1
β	2	α	1
β	2	β	1
β	2	β	2
β	2	γ	1

The Cartesian product we have discussed at depth. So, we assume for Cartesian product in relational algebra that the attributes are disjoint, that is the same attribute does not occur to both relation. If they are not disjoint, then we will use renaming to make them disjoint in SQL we made those implicit by as and so on. But in algebra we will simply assume that they are not having the same name if they are there are some attributes which have common name which are common between r and s will make them distinct.

(Refer Slide Time: 9:18)

IIT Madras
Using Chrome

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Rename

Division

Module Summary

Rename Operation

- Allows us to name, and therefore to refer to, the results of relational-a
- Allows us to refer to a relation by more than one name.
- Example:

$$\rho_X(E)$$

returns the expression E under the name X
- If a relational-algebra expression E has arity n , then

$$\rho_X(A_1, A_2, \dots, A_n)(E)$$

returns the result of expression E under the name X , and with the att

So, we need a way to rename. So, if we have renamed is the row operator, if we have a expression relational algebraic expression E, so it is a relation. So, I can give it a new name which is x and these are the names of the attributes. Now, it is a selective choice, I can rename everything I can rename part of it, but obviously what needs to be maintained is if I am using n attributes here, then it means that what E produces is an unary result.

If I have, E has 5 attributes, and here I am giving 3 attributes or 6 attributes, that is not going to work. So that the returns the result of E under the name x with the attributes renamed to A1 to An. So, this renaming is kind of a completeness process for the relational algebra, so that I can in any situation, I should be able to compute the Cartesian product. Now, obviously, there could be other extensional operations also.

(Refer Slide Time: 10:54)

Division Operation

- The division operation is applied to two relations
- $R(Z) \div S(X)$, where $X \subset Z$. Let $Y = Z - X$ (and hence $Z = X \cup Y$); that is, attributes of R that are not attributes of S
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear, and with
 - $t_R[X] = t_S$ for every tuple t_S in S.
- For a tuple t to appear in the result T of the DIVISION, the values in t must appear in combination with every tuple in S
- Division is a derived operation and can be expressed in terms of other operations
- $r \div s \equiv \Pi_{R-S}(r) - \Pi_{R-S}(r) \setminus ((\Pi_{R-S}(r) \times s) \setminus \Pi_{R-S,S}(r))$

So, there is one operation, which we did not talk of, in terms of SQL. We talked of, for example, the other extensional operator we talked of was joint. So, we will talk about joint. But these are kind of derived operators, the operator that I am going to discuss now is known as the division operator. So, it is a very interesting operator. So, let me first take a couple of examples on division and then come back to this result, this formulation.

(Refer Slide Time: 11:37)



Division Examples

Module 16
Partha Pratim Das
Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Union
Difference
Intersection
Cartesian Product
Joins
Division
Module Summary

• R

Lecturer	Module	Subject
Brown	Compilers	
Brown	Databases	
Green	Prolog	Prolog
Green	Databases	
Lewis	Prolog	
Smith	Databases	

S

R | S

So what I am trying to do is divide one relation say, the relation r here with relation s . The meaning of that division is now naturally, the division is possible, if the attribute or attributes that you have in the s are also present in r . So you divide based on those attributes. So here that correspondence is between module and subject, assume that they are the same category.

Now what you do is, like, what does division do, say if I divide 15 by 3? I find out how many times 15 goes in 3. Say, so the result is 5, so it is a 3, 3, 3, 3, 3 5 times I do, I get 15, similar concept. So, this dividing relation is a table. I want to find the existence of that table on the corresponding attribute. So, that the values on the remaining attributes, the values on the r - s attributes are same. That is dividing.

So, here it is pathological, because it is only one entry is there. So, this table is prolog. So, I get a prolog here, I get a prolog here. Now, since there is only other attributes, this is R - S , is Lecturer, there is only one other attribute. So, obviously, it will be the same. So, my result is green, my quotient is green, and Lewis, green and Lewis. So let's, you know, add one more tuple here and see what happens.

(Refer Slide Time: 14:06)

Division Examples (2)

Module 16
Partha Pratim Das
Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Join
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

R		S	R / S
Lecturer	Module	Subject	Lecturer
1 Brown	Compilers	Databases	Green
2 Brown	Databases	Prolog	
3 Green	Prolog		
4 Green	Databases		
5 Lewis	Prolog		
6 Smith	Databases		

Handwritten annotations:
 - Blue arrows from 'R-S' label point to the 'Lecturer' and 'Module' columns of table R.
 - Blue arrows from 'S' label point to the 'Subject' column of table S.
 - Blue arrows from 'R/S' label point to the 'Lecturer' column of the result table.
 - Blue arrows from the 'Databases' and 'Prolog' rows in table S point to the 'Green' result in the 'R/S' table.
 - Blue arrows from the 'Green' rows in table R point to the 'Green' result in the 'R/S' table.
 - Blue arrows from the 'Prolog' row in table R point to the 'Green' result in the 'R/S' table.
 - Blue arrows from the 'Databases' row in table R point to the 'Green' result in the 'R/S' table.
 - Blue arrows from the 'Prolog' row in table R point to the 'Green' result in the 'R/S' table.
 - Blue arrows from the 'Databases' row in table R point to the 'Green' result in the 'R/S' table.

So, now I have this is corresponding this is R-S. Now, we have a table, which is databases and prolog. Remember the condition I am dividing, so I need two things, one is on S attributes, I must get this instance. And on R-S attribute, I must have same value. So, I have two tuples here. So, what do I see them? I see them let's if we, if we numbered these tuples 2, 3, 4, 5, 6 where do we see them 2 and 3, 2 and 5, 2 and 3, 2 and 5, 4 and 3, 4 and 5.

These are the possible pairs, which are it could be considered the instances of S. Now, if you see 3 and 2 and 3, 2 and 3, I have this table, but these values are different, R-S values are not same brown and green. So, 2, 3 is not acceptable. If I take 2 and 5, I have database prolog. So I get a subset of, do not look at consecutivity because you know, ordering is immaterial.

So, I've to get a value databases here, I get a value prolog here, but this is brown, this is Lewis not same will not work. Let's say 4 and 3, 4 and 3. Database, prolog, green and green, great. So, this will work. Let's take 4 and 5 database, prolog, green and Lewis this will not work. So, what finally survived is 4 and 3, 4 and 3. So, green is the result. This is this indeed is a division operator. So, if you think carefully, what does the division tell me is green is associated with all instances in S, green is associated with all instances in S. So, it is basically a for all kind of sense that comes in here.

(Refer Slide Time: 17:17)

Slide 1: Division Examples (3)

A	sno	pno
	s1	p1
	s1	p2
	s1	p3
	s1	p4
	s2	p1
	s2	p2
	s3	p2
	s4	p2
	s4	p4

B1	pno
	p2

A/B1

B2	pno
	p2
	p4

A/B2

B3	pno
	p1

A/B3

Slide 2: Division Examples (3)

A	sno	pno
	s1	p1
	s1	p2
	s1	p3
	s1	p4
	s2	p1
	s2	p2
	s3	p2
	s4	p2
	s4	p4

B1	pno
	p2

A/B1

B2	pno
	p2
	p4

A/B2

B3	pno
	p1

A/B3

Let's take some more examples. I will go back to the formal definition. Do not worry. So, here is a table A. So, these are the Bs, this is A-Bs and there are the different instances of B. So, one is p1, p2. p2 is here, p2 is here, p2 is here, p2 is here, there is only one other attribute and here is only one as well. So, all these four are candidates, so they go in the output.

Now, consider p2, p4. So, you have p2 here, you have p4 here, p2 here, p2 here, p2 here, p4 here. We have to see which p2 p4 pair has the same value, this p2 p4 has s1. This p2, p4 does not work, this p2, p4 does not work, this p2 this p2, p4 works s4. So, we have s1 and s4 as I divide A by B2.

If we divide A by B3 p1, p2, p4, p1, p2, p4, p1, p2, p2, p2, p4. So, you can clearly see p1, p2, p4 all of them has s1. You take p1 with p4, anywhere else, you do not have the same value. So, p1 p2 p4 cannot have the same value. So, in this case the result is s1. So that is that is the kind of the semantics of division.

(Refer Slide Time: 19:28)

Division Example (4)

• Relations r, s :

A	B
α	1
α	2
α	3
β	1
γ	1
δ	1
δ	3
δ	4
ϵ	6
ϵ	1
β	2

r

e.g. A is customer name
B is branch name

So, let us say we have this relation, A and B, and these are the values and say A is a customer name and B is a branch name. So, 1, 2, I have 1, 2 here where there is alpha and I have 1 and 2 here, where the value is beta all other 1, 2 pairs are different. So, if this is a customer name and if this is branch, branch name, this is a customer name and this is a branch name, then what does it mean?

That this customer alpha has an account in every branch. Similarly, the customer beta has an account in every branch. So, if we asked to find customers who have an account in all branches of the bank, then I can simply take the branches table and divide the account holding table by that whatever is a quotient are those set of customers. That is the power of the division operator.

(Refer Slide Time: 20:50)

IIT Madras
1959

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Joins

Division

Module Summary

Division Example (5)

• Relations r, s :

A	B	C	D	E
a	a	a	a	1
a	a	y	a	1
a	a	y	b	1
β	a	y	a	1
β	a	y	b	3
y	a	y	a	1
y	a	y	b	1
y	a	β	b	1

r

A	B	C
a	a	y
v	a	v

D
a
b

s

e.g. Students who have courses, with inst

(Find students who hav
hav insts)

This example is similar just to show you multiple stuff, students who have taken both A and B courses with instructor 1, find students who have taken all courses given by instructor 1. So, what we are doing with A and so, D E, these are the columns on which I am doing the division and looking for A1 B1. I have A1 here, A1 here, A1 here.

Now, what A1 B1 matches in the rest is this one. Similarly, here at this one, no A1 B1 matches with the others. So, this is the result of this division, which gives me the students who have taken all courses that are offered by instructor 1. So, this is the, I mean I particularly work through a number of examples here, because I feel division is a little bit more complex operation and it is very useful.

(Refer Slide Time: 22:21)



IIT Madras
1959

Division Operation

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Join

Division

Module Summary

- The division operation is applied to two relations
- $R(Z) \div S(X)$, where $X \subseteq Z$. Let $Y = Z - X$ (and hence $Z = X \cup Y$); that is, attributes of R that are not attributes of S
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear in R with t , and with
 - $t_R[X] = t_S$ for every tuple t_S in S .
- For a tuple t to appear in the result T of the DIVISION, the values in t must appear in combination with every tuple in S
- Division is a derived operation and can be expressed in terms of other operations
- $$r \div s \equiv \Pi_{R-S}(r) - \Pi_{R-S}(r) \left(\left(\Pi_{R-S}(r) \times s \right) - \Pi_{R-S,S}(r) \right)$$



IIT Madras
1959

Division Operation

Module 16

Partha Pratim Das

Week Recap

Objectives & Outline

Relational Algebra

Select

Project

Union

Difference

Intersection

Cartesian Product

Join

Division

Module Summary

- The division operation is applied to two relations
- $R(Z) \div S(X)$, where $X \subseteq Z$. Let $Y = Z - X$ (and hence $Z = X \cup Y$); that is, attributes of R that are not attributes of S
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear in R with t , and with
 - $t_R[X] = t_S$ for every tuple t_S in S .
- For a tuple t to appear in the result T of the DIVISION, the values in t must appear in combination with every tuple in S
- Division is a derived operation and can be expressed in terms of other operations
- $$r \div s \equiv \Pi_{R-S}(r) - \Pi_{R-S}(r) \left(\left(\Pi_{R-S}(r) \times s \right) - \Pi_{R-S,S}(r) \right)$$

So, let me get back to the basic definition is if we have a set of attributes, Z and X and let Y be difference of $Z-X$. So, Y will be the attributes of the quotient, the result of the relation T that includes a tuple t_R appear in R with $t_R[Y]=t$ and so, you have a tuple t , which is got Y attributes, got X attributes.

So, this part is called $t[Y]$, where Y is the total minus X , $Z-X$ and this is called $t[X]$. So, we are saying that we will take that $t_R[Y]$ where $t_R[X]$ is equal to t_S for every tuple in S , the whole table. This part of the whole table has to come that is a formal definition of this and you can

check out for a tuple to appear in result t of the division the value t must appear in R in combination with every tuple in S .

So, now it will not be difficult to work out that division is not an independent operation, but you can actually get the division by using the other relational algebra operators that you have. So, you first compute this which is $R \div S$ of r which means the result table, the possibilities of the result table and these are possibilities of the result table you cannot have a result which is outside of this, this part.

And you take a product with s . So, you are generating all possible combinations, not all of them are there. Then you take out $R \div S$ of r . So, you take out which exists. So, then you project that on the $R \div S$ remaining and subtract from the origin which will give you the. So, if you have difficulty in understanding this expression, I would suggest that you work through a given example step by step and you will be able to understand. If after that you will still do not understand do let us know we will discuss, but this is division is a very very powerful operation to be done besides the four I mean six basic relational algebra operators and division.

(Refer Slide Time: 25:58)

Module Summary

Module 16
Partha Pratim Das

- Discussed relational algebra with examples

Week Recap
Objectives & Outline
Relational Algebra
Select
Project
Join
Difference
Intersection
Cartesian Product
Rename
Division
Module Summary

$R \bowtie_{\theta} (r, s)$
 $= \sigma_{\theta}(r \times s)$

Slides used in this presentation are borrowed from <http://db-book> permission of the authors.
Edited and new slides are marked with "PPD".

Obviously, we have the join which you already know written in this form where you can, you actually do a cross product and then you evaluate a condition on the attributes which is called a theta predicate of the join. Now, this theta predicate could be equality of common attributes on two tables then it becomes a natural join.

So, there are varieties of joint which can be written in this form. It is really not theta, I am sorry. So, join theta r, s is basically you take the Cartesian product and you do the selection based on the theta condition. So, this is called theta join, if theta is equality, it is called equal join. If theta enforces the values on the identical columns between r and s to be equal it is called natural join and so on so forth.

So, all of them can actually be written in terms of the selection and cross product and therefore, we do not again say that joint is an independent operation, but it is more like a derived operation. So, we conclude here on this module, so, we have taken a little bit in depth look into the relational algebra with some examples. Thank you very much for your attention and see you in the next module.

